

Marcus W. Fedarko

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SUMMARY

I'm a postdoc at the University of Maryland. I develop computational methods to help people study microbiome sequencing data in high resolution, with foci in metagenome assembly and data visualization.

EDUCATION

Ph.D., Computer Science UNIVERSITY OF CALIFORNIA, SAN DIEGO	9/2018–6/2025 LA JOLLA, CA
M.S., Computer Science UNIVERSITY OF CALIFORNIA, SAN DIEGO	9/2018–6/2022 LA JOLLA, CA
B.S. with High Honors, Computer Science UNIVERSITY OF MARYLAND	9/2014–5/2018 COLLEGE PARK, MD

RESEARCH EXPERIENCE

Postdoctoral Associate UNIVERSITY OF MARYLAND	10/2025–Present COLLEGE PARK, MD
Graduate Student Researcher UNIVERSITY OF CALIFORNIA, SAN DIEGO	9/2018–10/2025 LA JOLLA, CA
Research Intern UNIVERSITY OF MARYLAND	6/2016–8/2018 COLLEGE PARK, MD

TEACHING EXPERIENCE

Graduate Teaching Assistant UNIVERSITY OF CALIFORNIA, SAN DIEGO	LA JOLLA, CA
• CSE 185: Advanced Bioinformatics Laboratory	3/2025–6/2025
• CSE 181: Molecular Sequence Analysis	1/2025–3/2025 1/2024–3/2024
• CSE 282: Introduction to Bioinformatics Algorithms	1/2025–3/2025 1/2023–3/2023 1/2022–3/2022 1/2021–3/2021
Course Assistant MARINE BIOLOGICAL LABORATORY	WOODS HOLE, MA
• STAMPS: Strategies and Techniques for Analyzing Microbial Population Structures	7/2018–8/2018
• MOLE: Workshop on Molecular Evolution	7/2018
Undergraduate Teaching Assistant UNIVERSITY OF MARYLAND	COLLEGE PARK, MD

OTHER PROGRAMMING / WRITING EXPERIENCE

Software Developer <i>Bioinformatics Algorithms</i> (COMPEAU & PEVZNER)	7/2025–10/2025 REMOTE
Student Staff Writer UNIVERSITY OF MARYLAND DEPARTMENT OF COMPUTER SCIENCE	1/2015–9/2017 COLLEGE PARK, MD
Student Intern AXIOMETRIC	5/2013–8/2014 COLUMBIA, MD
Intern Software Engineer BATTLEFIELD TELECOMMUNICATIONS SYSTEMS	7/2012–8/2012 COLUMBIA, MD

PEER-REVIEWED PAPERS

8. Zhang Z, **Fedarko MW**, Bankevich A, and Pevzner PA (2026). “MGA: a tool for haplotype-mixed assembly of long and accurate reads.” *Genome Biology*, 27(1):207.
7. **Fedarko MW**, Kolmogorov M, and Pevzner PA (2022). “Analyzing rare mutations in metagenomes assembled using long and accurate reads.” *Genome Research*, 32(11-12):2119–2133.
6. Cantrell K*, **Fedarko MW***, Rahman G, McDonald D, Yang Y, Zaw T, Gonzalez A, Janssen S, Estaki M, Haiminen N, Beck KL, Zhu Q, Sayyari E, Morton JT, Armstrong G, Tripathi A, Gauglitz JM, Marotz C, Matteson NL, Martino C, Sanders JG, Carrieri AP, Song SJ, Swafford AD, Dorrestein PC, Andersen KG, Parida L, Kim H-C, Vázquez-Baeza Y, and Knight R (2021). “EMPress Enables Tree-Guided, Interactive, and Exploratory Analyses of Multi-omic Data Sets.” *mSystems*, 6(2):e01216-20. (* = contributed equally)
5. Huey SL, Jiang L, **Fedarko MW**, McDonald D, Martino C, Ali F, Russell DG, Udipi SA, Thorat A, Thakker V, Ghugre P, Potdar RD, Chopra H, Rajagopalan K, Haas JD, Finkelstein JL, Knight R, and Mehta S (2020). “Nutrition and the Gut Microbiota in 10- to 18-Month-Old Children Living in Urban Slums of Mumbai, India.” *mSphere*, 5(5):e00731-20.
4. **Fedarko MW**, Martino C, Morton JT, González A, Rahman G, Marotz CA, Minich JJ, Allen EA, and Knight R (2020). “Visualizing ’omic feature rankings and log-ratios using Qurro.” *NAR Genomics and Bioinformatics*, 2(2):lqaa023.
3. Sanders JG, Nurk S, Salido RA, Minich J, Xu ZZ, Martino C, **Fedarko M**, Arthur TD, Chen F, Boland BS, Humphrey GC, Brennan C, Sanders K, Gaffney J, Jepsen K, Khosroheidari M, Green C, Liyange M, Dang JW, Phelan VV, Quinn RA, Bankevich A, Chang JT, Rana TM, Conrad DJ, Sandborn WJ, Smarr L, Dorrestein PC, Pevzner PA, and Knight R (2019). “Optimizing sequencing protocols for leaderboard metagenomics by combining long and short reads.” *Genome Biology*, 20(1):226.
2. Ghurye J, Treangen T, **Fedarko M**, Hervey WJ, and Pop M (2019). “MetaCarvel: linking assembly graph motifs to biological variants.” *Genome Biology*, 20(1):174.
1. Meisel JS, Nasko DJ, Brubach B, Cepeda-Espinoza V, Chopyk J, Corrada-Bravo H, **Fedarko M**, Ghurye J, Javkar K, Olson ND, Shah N, Allard SM, Bazinet AL, Bergman NH, Brown A, Caporaso JG, Conlan S, DiRuggiero J, Forry SP, Hasan NA, Kralj J, Luethy PM, Milton DK, Ondov BD, Preheim S, Ratnayake S, Rogers SM, Rosovitz MJ, Sakowski EG, Schliebs NO, Sommer DD, Ternus KL, Uritskiy G, Zhang SX, Pop M, and Treangen TJ (2018). “Current progress and future opportunities in applications of bioinformatics for biodefense and pathogen detection: Report from the Winter Mid-Atlantic Microbiome Meet-up, College Park, MD January 10th, 2018.” *Microbiome*,

OPEN-SOURCE SOFTWARE

6. **wotplot**: Library for creating and visualizing dot plot matrices.
<https://github.com/fedarko/wotplot>
5. **strainFlye** (🔥₇): Pipeline for the analysis of rare mutations in metagenomes.
<https://github.com/fedarko/strainFlye>
4. **EMPress** (🔥₆): Visualization tool for phylogenetic trees and associated data.
<https://github.com/biocore/empress>
3. **Qurro** (🔥₄): Visualization tool for log-ratios of compositional data.
<https://github.com/biocore/qurro>
2. **pyfastg**: Library for parsing FASTG-format assembly graph files.
<https://github.com/fedarko/pyfastg>
1. **MetagenomeScope**: Visualization tool for metagenome assembly graphs.
<https://github.com/marbl/MetagenomeScope>

Projects marked (🔥_{*n*}) represent the “main contribution” of the *n*-th peer-reviewed paper in the list above.

TALKS

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| 7. “Hierarchical decomposition and visualization of metagenome assembly graphs.”
<i>CAVE (Chesapeake Aquatic Viral Ecology) 2026.</i> | 4/2026 |
| 6. “Hierarchical decomposition and visualization of metagenome assembly graphs.”
<i>M³ (Mid-Atlantic Microbiome Meet-up) 2026.</i> | 3/2026 |
| 5. “Metagenome assembly.”
<i>Guest lecture for ESE 184 (Computational Tools for Decoding Microbial Ecosystems), California Institute of Technology.</i> | 1/2024 |
| 4. “Studying microbiomes using DNA sequencing.”
<i>Presentation to students from Kearny High School visiting UC San Diego.</i> | 5/2023 |
| 3. “EMPress: a tool for tree-guided exploratory analysis of multi-omic datasets.”
<i>UC San Diego CMI Connect Webinar.</i> | 1/2021 |
| 2. “Visualizing, exploring, and understanding microbiome sequencing data.”
<i>UC San Diego CSE Research Open House.</i> | 1/2020 |
| 1. “Visualizing metagenomic assembly graphs, doing undergrad research at UMD, applying to grad schools, and probably other stuff along the way.”
<i>Guest lecture for CMSC 396H (Undergraduate Honors Seminar), University of Maryland.</i> | 4/2018 |

EDITING AND REVIEWING

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| 2. Textbook Editor : <i>Advanced Bioinformatics Algorithms</i> | 2024 |
| 1. Peer Reviewer : <i>Bioinformatics Advances, PLOS ONE</i> | Ad hoc |

SERVICE

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| 8. Volunteer : Maryland Day “Bites and Bytes” booth | 4/2026 |
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7. Organizer: “Algorithmic Bioinformatics” seminar series	12/2025–
6. Panelist: CSE Society at UCSD × Triton Software Engineering research panel	5/2025
5. System Administrator: Pevzner Lab computing server	2/2023–10/2025
4. Mentor: UC San Diego Graduate Women in Computing mentorship program	10/2021–6/2024
3. Moderator: QIIME 2 forum (https://forum.qiime2.org)	3/2020–4/2024
2. Co-organizer: UC San Diego CSE Visit Day	1/2019–3/2024
1. Code Review (Co-)organizer: Knight Lab	12/2018–8/2020

HONORS AND AWARDS

10. Chez Bob Hackathon Award: UC San Diego	2024
9. CMNS Dean’s List: University of Maryland	2014–2018
8. University Honors Citation: University of Maryland	2017
7. Rita Colwell Travel Fellowship: University of Maryland	2017
6. Travel Award, EGS Workshop: University of Michigan	2017
5. John D. Gannon Endowed Scholarship: University of Maryland	2017
4. Corporate Partners in Computing Scholarship: University of Maryland	2016, 2017
3. Member: Omicron Delta Kappa National Leadership Honor Society	2016
2. Scholarship for Employees’ Children: Northrop Grumman	2014
1. Dean’s Scholarship: University of Maryland	2014

SKILLS

- **Languages:** English (native), German (basic), Hindi (basic)
- **Computer Languages:** Python, JavaScript, HTML / CSS, L^AT_EX, Java, C, C++, ...
- **Software Libraries:** matplotlib, pandas, NumPy, SciPy, scikit-bio, Dash, Cytoscape.js, ...